

TITLE STATUS UPDATE (TSU) EXTENSION TO THE KLAES VIRTUAL FILE SYSTEM (VFS) FRAMEWORK

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1. Overview & Conceptual Distinction

While the SPU/APU (DPU) framework governs the **spatial and structural transformation** of parcels (birth, subdivision, merger, and rebirth of FileNos), the Title Status Update (TSU) framework governs the **legal and administrative lifespan** of those parcels.

A parcel may undergo multiple SPU/APU transformations throughout its life, but at any point, its legal existence can be truncated, suspended, modified, or terminated by a TSU event. Unlike SPU/APU, which creates new Logical Folios, TSU primarily alters the state of an existing Logical Folio — though in some cases (e.g., Amendment), it may spawn a derivative folio.

The critical design principle remains the same: the physical EDMS folder structure is never altered. All status changes are managed through the backend control layer, and the frontend Virtual File Explorer dynamically reflects the current legal standing of each folio.

2. The Six TSU Event Types

SN	TSU Type	Trigger Authority	Effect on Folio	Reversible?	Creates New Folio?
1	Litigation	Court Order	Suspends all transactions on the folio; freezes the file	☑ Yes (upon court resolution)	✗ No
2	Cancellation	Ministry (Director of Lands)	Terminates the application or title; marks folio as void	⚠ Rarely (via appeal/re-grant)	✗ No
3	Withdrawal	Applicant (Application) or Ministry (Title)	Removes the application from processing or withdraws an issued title	⚠ Conditional (re-application possible)	✗ No
4	Revocation	Commissioner or Governor	Extinguishes the title under the Land Use Act; most severe action	⚠ Rarely (via judicial review)	✗ No
5	Amendment	Ministry (Director of Lands)	Modifies specific attributes of the title (name, area, purpose) without extinguishing it	✗ N/A (superseded by new amendment)	⚠ Optionally (derivative folio)
6	Closure	Ministry / System Admin	Administratively closes a dormant, duplicate, or erroneous file	☑ Yes (re-opening possible)	✗ No

3. Backend Database Architecture Extension

A. New Core Tables

1. `Title\_Status\_Updates` (Primary TSU Event Log)

This table captures every TSU event that occurs against a Logical Folio.

SN	Column	Data Type	Description
1	tsu_id	UUID (PK)	Unique identifier for the TSU event
2	folio_id	UUID (FK → Logical_Folios)	The folio being affected
3	tsu_type	ENUM	LITIGATION", "CANCELLATION", "WITHDRAWAL", "REVOCATION", "AMENDMENT", "CLOSURE
4	tsu_subtype	VARCHAR	Granular classification (e.g., WITHDRAWAL_BY_APPLICANT", "WITHDRAWAL_BY_MINISTRY", "REVOCATION_BY_GOVERNOR", "REVOCATION_BY_COMMISSIONER)
5	trigger_authority	VARCHAR	Name/designation of the authority (e.g., "Hon. Justice A.B. Ibrahim", "Executive Governor of Kano State", "Director of Lands")
6	reference_number	VARCHAR	Court case number, Ministry memo reference, or Gazette number
7	effective_date	DATE	The date the TSU takes legal effect
8	expiry_date	DATE (Nullable)	For temporary statuses like Litigation (resolved date) or Closure (re-opened date)
9	is_active	BOOLEAN	Whether this TSU is currently in force
10	is_reversed	BOOLEAN	Whether this TSU has been overturned
11	reversal_tsu_id	UUID (FK → self, Nullable)	Links to the TSU that reversed this one
12	justification	TEXT	Detailed reason for the TSU
13	created_by	UUID (FK → Users)	The KLAES user who recorded the TSU
14	created_at	TIMESTAMP	System timestamp

2. `Folio\_Status\_History` (Chronological State Machine)

This table tracks every status transition a folio undergoes, creating an unbreakable audit trail. It works in tandem with the existing `Parcel\_Transactions` table but focuses exclusively on legal status rather than spatial transformation.

SN	Column	Data Type	Description
1	history_id	UUID (PK)	Unique identifier
2	folio_id	UUID (FK → Logical_Folios)	The folio
3	previous_status	ENUM	The status before this change (e.g., ACTIVE", "LITIGATION_HOLD)
4	new_status	ENUM	The status after this change
5	trigger_event_type	ENUM	SPU", "APU" or "TSU
6	trigger_event_id	UUID	FK to either Parcel_Transactions.transaction_id or Title_Status_Updates.tsu_id

7	changed_by	UUID (FK → Users)	
8	changed_at	TIMESTAMP	
9	remarks	TEXT	

3. `Litigation\_Cases` (Court-Specific Details)

Since Litigation is the most complex TSU type (involving external judicial processes), it warrants a dedicated table.

SN	Column	Data Type	Description
1	case_id	UUID (PK)	
2	tsu_id	UUID (FK → Title_Status_Updates)	Links to the parent TSU event
3	court_name	VARCHAR	e.g., "Kano State High Court", "Federal High Court Abuja"
4	case_number	VARCHAR	e.g., "K/123/2026"
5	plaintiff_name	VARCHAR	
6	defendant_name	VARCHAR	
7	filing_date	DATE	
8	judgment_date	DATE (Nullable)	
9	judgment_summary	TEXT	
10	injunction_type	ENUM	INTERIM, "INTERLOCUTORY", "PERPETUAL", "NONE"
11	affected_folios	JSON	Array of folio_ids if the litigation affects multiple parcels
12	status	ENUM	PENDING, "RESOLVED", "DISMISSED", "APPEALED"

4. `TSU\_Supporting\_Documents` (Virtual Attachment Layer)

Instead of physically moving court orders or revocation notices into the EDMS folder, this table maps them to the folio virtually.

SN	Column	Data Type	Description
1	doc_id	UUID (PK)	
2	tsu_id	UUID (FK → Title_Status_Updates)	
3	physical_file_id	UUID (FK → Physical_Files)	The EDMS folder where the document physically resides
4	page_index	INTEGER	The specific page/image
5	document_type	ENUM	COURT_ORDER, "REVOCATION_NOTICE", "GAZETTE_NOTICE", "WITHDRAWAL_LETTER", "AMENDMENT_APPROVAL", "CLOSURE_MEMO", "MINISTRY_LETTER"
6	uploaded_by	UUID (FK → Users)	

7	uploaded_at	TIMESTAMP	
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B. Updates to Existing Tables

`Logical\_Folios` Table — New Columns

SN	Column	Data Type	Description
1	current_tsu_status	ENUM	NONE, LITIGATION_HOLD, CANCELLED, WITHDRAWN, REVOKED, AMENDED, CLOSED
2	tsu_status_since	DATE	When the current TSU status was applied
3	is_transaction_frozen	BOOLEAN	If TRUE, no SPU/APU or new RofO can be processed on this folio
4	freeze_reason	VARCHAR	e.g., "Active Litigation — Court Order K/123/2026"

4. TSU Processing Workflows

Workflow 1: Litigation (Court Order)

1. A court order is received at the Land Registry.
2. The Land Officer creates a new `Title\_Status\_Updates` record (Type: `LITIGATION`).
3. The system automatically creates a `Litigation\_Cases` record with court details.
4. The `Logical\_Folios.current\_tsu\_status` is set to `LITIGATION\_HOLD`.
5. The `Logical\_Folios.is\_transaction\_frozen` flag is set to `TRUE`.
6. Frontend Effect: The Virtual File Explorer displays a red banner across the folio: " LITIGATION HOLD — All transactions suspended per Court Order K/123/2026." Any attempt to initiate an SPU/APU on this folio is blocked by the system.
7. Resolution: When the court resolves the matter, a new TSU record is created (Type: `LITIGATION`, Subtype: `RESOLUTION`), the `is\_reversed` flag on the original TSU is set to `TRUE`, and the folio's freeze is lifted.

Workflow 2: Cancellation (Ministry)

1. The Director of Lands issues a cancellation memo.
2. A `Title\_Status\_Updates` record is created (Type: `CANCELLATION`).
3. The supporting memo is scanned into the EDMS (into the existing physical folder) and mapped via `TSU\_Supporting\_Documents`.

4. The folio status changes to `CANCELLED`.
5. Frontend Effect: The Virtual File Explorer displays a dark grey overlay with a watermark: "CANCELLED." The file contents remain viewable for historical/audit purposes, but no new transactions can be initiated.
6. Important: Cancellation does NOT delete the physical EDMS folder or the Logical Folio record. It merely changes the legal status. The full history remains intact.

Workflow 3: Withdrawal

1. Applicant Withdrawal: The applicant submits a withdrawal letter. A TSU record is created (Subtype: `WITHDRAWAL\_BY\_APPLICANT`). The folio status changes to `WITHDRAWN`.
2. Ministry Withdrawal: The Ministry withdraws an erroneously issued title. A TSU record is created (Subtype: `WITHDRAWAL\_BY\_MINISTRY`). The folio status changes to `WITHDRAWN`.
3. Frontend Effect: An amber banner appears: `***WITHDRAWN — [Applicant/Ministry] withdrawal effective [date].**`
4. If the applicant re-applies in the future, a new Logical Folio is created, and the `Folio\_Lineage\_Mapping` table links the new folio to the withdrawn one for traceability.

Workflow 4: Revocation (Commissioner/Governor)

1. A revocation notice is published in the Official Gazette.
2. A `Title\_Status\_Updates` record is created (Type: `REVOCATION`, Subtype: `REVOCATION\_BY\_GOVERNOR` or `REVOCATION\_BY\_COMMISSIONER`).
3. The Gazette reference number is recorded in `reference\_number`.
4. The folio status changes to `REVOKED`. This is the most severe TSU — it extinguishes the title under Section 28 of the Land Use Act.
5. Frontend Effect: A bold red banner with a lock icon: " REVOKED — Title extinguished per Gazette Notice No. XX of 2026 by the Executive Governor of Kano State." The file is locked in read-only mode. All linked SPU/APU child folios are flagged for review.
6. Cascade Effect: If the revoked folio was a parent in a previous Subdivision, the system alerts the Land Officer that all child folios derived from the revoked parent may need legal review.

Workflow 5: Amendment

1. An amendment application is approved (e.g., correcting the owner's name, adjusting the plot area, or changing the land use without a full SPU CoP process).
2. A `Title\_Status\_Updates` record is created (Type: `AMENDMENT`).
3. Two sub-scenarios:
 - In-Place Amendment: The existing folio's attributes (name, area, purpose) are updated directly. The `Folio\_Status\_History` logs the before/after values. The folio status becomes `AMENDED` (but remains active and transactable).
 - Derivative Amendment: If the amendment is substantial enough to warrant a new FileNo, a new `Logical\_Folio` is created, and the `Folio\_Lineage\_Mapping` links the old folio (Parent) to the new folio (Child). The old folio is marked as `SUPERSEDED`.
4. Frontend Effect: A blue info banner: "AMENDED — [Attribute] changed from [Old Value] to [New Value] effective [date]." The Virtual File Explorer includes the amendment approval letter as a virtual document overlay.

Workflow 6: Closure

1. An administrative decision is made to close a file (e.g., duplicate file, abandoned application, or erroneous entry).
2. A `Title\_Status\_Updates` record is created (Type: `CLOSURE`).
3. The folio status changes to `CLOSED`.
4. Frontend Effect: A grey banner: "CLOSED — Administrative closure effective [date]. Reason: [justification]."
5. Re-opening: If the closure was erroneous, a new TSU record (Type: `CLOSURE`, Subtype: `REVERSAL`) is created, the original closure is marked `is\_reversed = TRUE`, and the folio reverts to `ACTIVE`.

5. Frontend UI Framework Extension

A. The Virtual File Explorer — TSU-Aware Display

When a user opens a folio in the KLAES Virtual File Explorer, the system now checks the `Logical\_Folios.current\_tsu\_status` field and renders the UI accordingly:

SN	TSU Status	Banner Color	Banner Text	File Accessibility	Transaction Lock
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1	NONE	None	—	Full access	None
2	LITIGATION_HOLD	● Red	"LITIGATION HOLD — Transactions Suspended"	Read-only	🔒 Full lock
3	CANCELLED	● Dark Grey	"CANCELLED — Title Void"	Read-only + Watermark	🔒 Full lock
4	WITHDRAWN	○ Amber	"WITHDRAWN"	Read-only	🔒 Full lock
5	REVOKED	● Bold Red	"REVOKED — Title Extinguished"	Read-only + Lock icon	🔒 Full lock + Cascade alert
6	AMENDED	○ Blue	"AMENDED — See History"	Full access	None
7	CLOSED	○ Grey	"CLOSED — Administrative"	Read-only	🔒 Full lock

B. The TSU Timeline Tab

In addition to the existing Lineage & History tab (which shows SPU/APU events), a new Status Timeline tab is added to the Virtual File Explorer:

1	📅	2019-03-15	✅	ALLOTMENT – Original title issued (FileNo: 1234)
2	📅	2021-07-22	🔗	SPU: SUBDIVISION – File 1234 → Files 1234A, 1234B
3	📅	2023-11-08	🔵	TSU: AMENDMENT – Owner name corrected (Jibril → Jibril Ibrahim)
4	📅	2025-01-14	🔴	TSU: LITIGATION – Court Order K/456/2025 (Frozen)
5	📅	2026-06-30	🟢	TSU: LITIGATION RESOLVED – Court dismissed case (Unfrozen)
6	📅	2026-08-20	🟣	TSU: CANCELLATION – Ministry cancelled title (Ref: MOL/2026/789)

Each node in the timeline is clickable, revealing the supporting documents (court orders, memos, Gazette notices) mapped via the `TSU\_Supporting\_Documents` table.

C. Permission & Role-Based Access Control (RBAC)

SN	Action	Land Officer	Director of Lands	Commissioner	System Admin
1	Record Litigation TSU	✅	✅	✅	❌
2	Record Cancellation TSU	❌	✅	✅	❌
3	Record Withdrawal TSU	✅ (Applicant) / ❌ (Ministry)	✅	✅	❌
4	Record Revocation TSU	❌	❌	✅	❌
5	Record Amendment TSU	✅	✅	✅	❌
6	Record Closure TSU	❌	✅	✅	✅

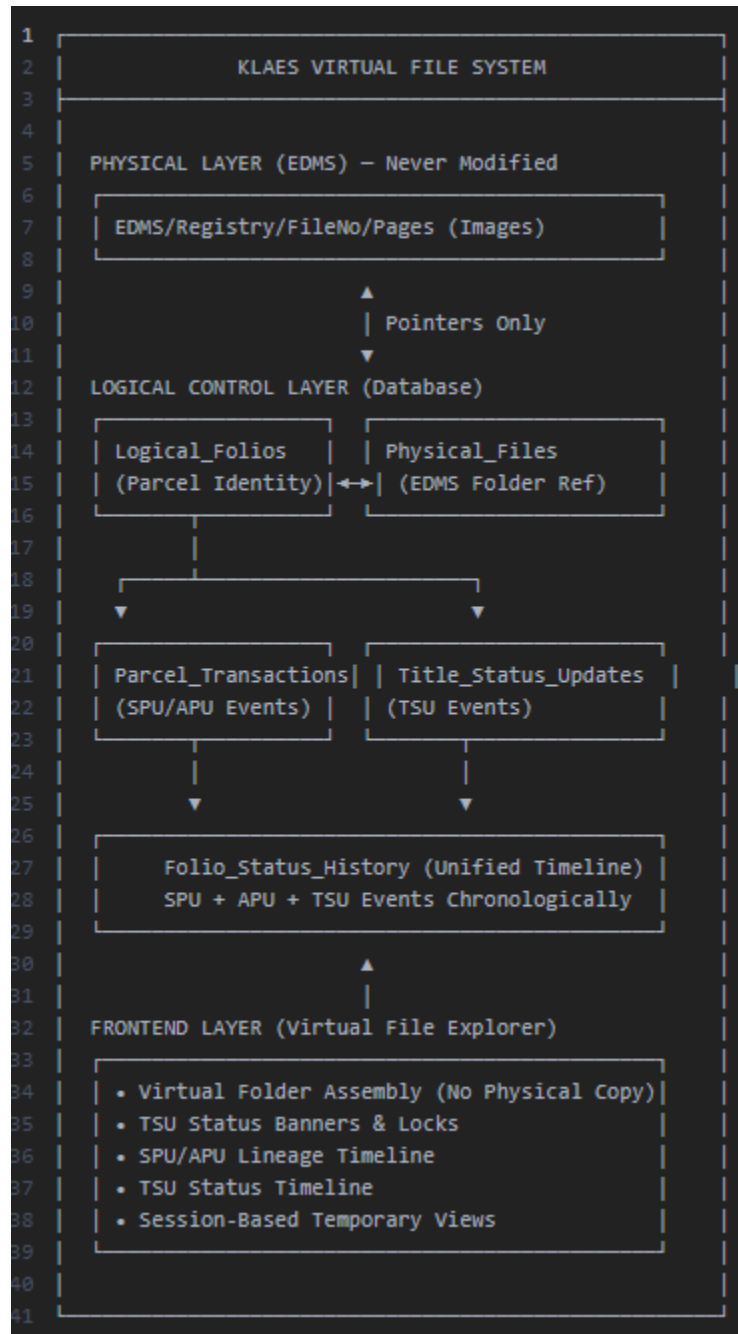
7	Reverse any TSU	✗	✗	✓	✓ (with audit override)
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6. Integration with the Existing SPU/APU Framework

The TSU framework is designed to coexist seamlessly with the SPU/APU framework through the unified `Folio\_Status\_History` table. This means:

1. A folio can have both SPU and TSU events in its timeline. For example, a folio may have been subdivided (SPU) in 2021, amended (TSU) in 2023, and then placed under litigation (TSU) in 2025. The timeline displays all events chronologically.
2. TSU freezes block SPU/APU processing. If a user attempts to initiate a Subdivision on a folio that is currently under `LITIGATION\_HOLD`, the system checks the `is\_transaction\_frozen` flag and blocks the operation with an error message.
3. Cascade alerts for Revocation. If a parent folio is revoked, the system queries the `Folio\_Lineage\_Mapping` table to identify all child folios derived from it (via Subdivision, Merger, etc.) and flags them for legal review. This is critical because if the parent title was extinguished, the legal basis for all derived child titles may be compromised.
4. Amendment as a bridge between TSU and SPU. An Amendment TSU that changes the land use of a folio is functionally similar to an SPU Change of Purpose. The system treats minor amendments (name corrections, area adjustments) as pure TSU events, but substantial amendments (land use changes, area expansions) trigger both a TSU record and an SPU record, ensuring full traceability.

7. Summary of the Unified VFS Architecture



8. Key Design Principles (Reiterated)

1. Zero Physical Duplication: TSU events never create, move, rename, or delete physical EDMS folders. All changes are logical.
2. Immutable Audit Trail: Every TSU event is permanently recorded. Even reversed or resolved statuses remain in the history for legal accountability.

3. Transaction Freezing: Active TSU statuses (Litigation, Cancellation, Revocation, Withdrawal, Closure) automatically freeze the folio against any new SPU/APU or RoFo processing.
4. Cascade Awareness: Revocation and Cancellation of parent folios trigger automatic alerts on all derived child folios.
5. Role-Based Governance: Only authorized personnel (Director, Commissioner, Governor) can initiate high-severity TSU events.
6. Reversibility: The framework supports the reversal of TSU events (e.g., litigation resolution, closure re-opening) through linked reversal records, maintaining a complete chain of custody.